

Editor-in-Chief

Dear Editor-in-Chief,

On behalf of my co-author and corresponding author, Professor Md. Jamal Uddin, I am pleased to resubmit our revised manuscript entitled:

“Association Between Internet Use and Unmet Need for Family Planning among Currently Pregnant Women in Bangladesh: Evidence from BDHS 2022” for reconsideration for publication in your esteemed journal.

We sincerely thank the Editor and reviewers for their valuable comments and constructive suggestions. We have carefully addressed all comments and revised the manuscript accordingly. We believe that these revisions have substantially improved the quality, clarity, and scientific contribution of our work.

A detailed point-by-point response to all reviewer comments is provided in the accompanying **Response to Reviewers** document. For transparency and ease of evaluation, we have submitted the following files:

- Clean revised manuscript (final version)
- Revised manuscript with highlighted changes (green-colored revisions)
- Response to Reviewers document (point-by-point responses)

In the revised manuscript with highlighted changes, all additions, modifications, and revisions are clearly marked in **green color**. Furthermore, **page numbers and line numbers corresponding to each revision have been provided in both the highlighted manuscript and the Response to Reviewers document**, enabling reviewers and editors to easily locate and verify the changes made in response to each comment.

Sincerely,

Md Salek Miah

On behalf of the Corresponding Author

Md. Jamal Uddin, PhD

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We sincerely thank the Editor and Reviewers for their careful evaluation and constructive feedback on our manuscript titled “*Association Between Internet Use and Unmet Need for Family Planning among Ever-Married Women in Bangladesh: Evidence from BDHS 2022.*”

REVIEWER 1

RESPONSE TO REVIEWER 1

RESPONSE TO REVIEWER 1 – COMMENT 1

Reviewer Comment:

Why do we need c-statistic? This study is on a causal framework, not in the predictive framework.

Author Response (Page 2, Lines 42-48)

We sincerely thank the reviewer for this important methodological clarification. In response to this comment, we have revised the manuscript by removing the c-statistic (AUC) from the Abstract and de-emphasizing its role throughout the manuscript. Model evaluation is now primarily based on measures more appropriate for an associational epidemiological framework, including likelihood ratio tests, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC), which were used to compare nested models and assess overall model fit.

The interpretation of findings now focuses on adjusted odds ratios, confidence intervals, and survey-weighted regression estimates rather than predictive discrimination metrics. These revisions improve the methodological consistency of the manuscript and better align the analysis with a causal inference framework.

Changes made:

The c-statistic (AUC) has been removed from the Abstract and de-emphasized throughout the manuscript. Model assessment has been revised to emphasize likelihood ratio tests, AIC, and BIC as the primary model evaluation approaches.

RESPONSE TO REVIEWER 1 – COMMENT 2

Reviewer Comment:

Please be specific. Showed? Is this prevalence? And unmet need for what?

Author Response (Page 2, Lines 42)

We sincerely thank the reviewer for this helpful comment. We agree that the original sentence was imprecise and did not clearly specify the outcome measure.

In response, we have revised the sentence to explicitly report the prevalence of unmet need for family planning among the study population. The revised sentence now reads:

“The prevalence of unmet need for family planning among currently pregnant women was 41.0%.”

This revision improves clarity, precision, and epidemiological reporting consistency.

Changes made:

The wording in the Results section has been revised to clearly report the prevalence of unmet need for family planning among currently pregnant women.

RESPONSE TO REVIEWER 1 – COMMENT 3

Reviewer Comment:

Again, be specific. E.g. 1.8 times higher odds compared to ...

Author Response (Page 2, Lines 43-44)

We sincerely thank the reviewer for this helpful suggestion. In response, we have revised the sentence to clearly identify the comparison category. The revised sentence now reads:

“Women who used the internet had 1.83 times higher odds of unmet need for family planning compared with women who did not use the internet (aOR = 1.83, 95% CI: 1.52–2.20).” This revision improves clarity and aligns the reporting with standard epidemiological interpretation of adjusted odds ratios.

Changes made:

The Results section has been revised to explicitly specify the reference category for the adjusted odds ratio.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

Very general sentence. Unmet need for what? Why do you need to state Africa? Is that necessary? Again, why do you need to mention that Asia has lower percentages? Remove this kind of generic statements. Revise the whole paper. Also, the temporal reference (2022 as today) is unclear.

Author Response (Page 3, Lines no: 61-68, and line no:78-85)

We sincerely thank the reviewer for this important and constructive feedback. We agree that the original paragraph included overly general global comparisons and unnecessary regional descriptions that did not directly strengthen the study rationale.

In response, we have substantially revised the paragraph to improve focus and clarity. We removed non-essential regional comparisons (e.g., Sub-Saharan Africa and Asia-specific contrasts) and eliminated generic statements that did not directly contribute to the study objective. The revised version now focuses specifically on unmet need for family planning and its relevance in Bangladesh using recent evidence from BDHS 2022.

We have also corrected temporal wording to ensure consistency and avoid implying “2022 as current time,” and instead refer appropriately to BDHS 2022 as the most recent nationally representative survey.

Changes made:

The entire paragraph has been rewritten to remove unnecessary global comparisons, improve specificity to family planning unmet need, and align temporal references appropriately with BDHS 2022 data.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

Remove this kind of generic statements. Revise the whole paper. 2026? The reference from Kundu was published in 2022. Why are you mentioning 2022 as today?

Author Response (Page 4, Lines 95-100)

We sincerely thank the reviewer for this careful and constructive feedback. We agree that the original introduction contained overly generic statements and unclear temporal phrasing, which may have reduced the scientific precision and clarity of the manuscript.

In response, we have comprehensively revised the entire Introduction section to improve specificity, relevance, and alignment with the study objective. All non-essential and overly general statements have been removed, and the revised version now focuses directly on unmet need for family planning, internet use, and the Bangladesh context.

We have also corrected all temporal references to ensure accuracy and consistency. In particular, we have removed any phrasing implying that 2022 represents the “present” or “today,” and instead clearly refer to BDHS 2022 as the most recent available nationally representative survey data. This ensures proper academic temporal framing and avoids misinterpretation.

Changes made:

The Introduction section has been fully revised to remove generic background statements, improve focus on study-specific rationale, and correct temporal references related to BDHS 2022, ensuring clarity and academic precision.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

You can insert these numbers in the diagram. It would be easier.

Author Response (Page 7, Figure 1, line no: 161-162)

We sincerely thank the reviewer for this helpful suggestion. In response, we have revised Figure 1 to incorporate the key sampling and participant selection numbers directly within the flow diagram. The updated figure now clearly presents the number of enumeration areas (EAs), interviewed participants, exclusions, missing observations, and the final analytical sample in a stepwise format. This revision improves clarity, transparency, and overall readability of the sampling procedure and analytic sample derivation.

Changes made:

Figure 1 has been revised to include detailed sampling and exclusion numbers directly within the participant flow diagram.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

For the first time I am getting 'pregnant' in this study. Why was it not mentioned in the title, abstract, or objectives? Poor writing quality. You need to mention it explicitly as we still do not know the definition of the outcome. Can there be a possible bias?

Author Response (Page 6, Lines 157-158, page no: 1, line no: 1-2, page no: 2, line no: 32, 36)

We sincerely thank the reviewer for this important observation. We agree that the restriction to currently pregnant women should have been explicitly stated in the title, abstract, and objectives to improve clarity and consistency. In response, we have revised these sections accordingly to clearly indicate that the study population consists of currently pregnant women aged 15–49 years.

We also acknowledge the reviewer's concern regarding potential bias arising from restricting the analysis to currently pregnant women. This restriction implies that the findings are specific to a selected subgroup of women who were pregnant at the time of the survey, and therefore may not be fully generalizable to all ever-married women or all reproductive-aged women in Bangladesh. However, this analytical restriction was necessary because unmet need in this study was

operationalized based on pregnancy intention status, which is only applicable to currently pregnant women within the DHS framework used in this analysis.

We have now explicitly clarified this as a study limitation in the revised manuscript to improve transparency regarding the scope and generalizability of the findings.

Changes made:

The title, abstract, and objectives have been revised to explicitly state the study population as currently pregnant women, and a clarification regarding the limited generalizability due to this analytical restriction has been added to the limitations section.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

I am not sure if this is the proper definition. See the DHS revised definition and diagram.

Author Response (Page 7-8, Lines 168-177)

We sincerely thank the reviewer for this important methodological observation and for referring us to the DHS revised unmet need framework. In response, we carefully revisited the DHS guidance and revised the outcome variable definition to better align with the DHS revised unmet need indicator (v626a) and its analytical framework.

The revised manuscript now explicitly clarifies that the study population was restricted to currently pregnant women and that unmet need status was operationalized based on the retrospective wantedness of the current pregnancy, consistent with the DHS framework for pregnant women. Pregnancies reported as mistimed or unwanted were classified as unmet need, whereas intended pregnancies were classified as no unmet need.

We have also clarified that this represents a pregnancy-intention-based application of the DHS revised unmet need framework adapted to the analytical design of the present study, rather than a full population-based estimation of unmet need used in standard DHS reports.

Changes made:

The outcome variable definition in the Methods section has been substantially revised to improve consistency with the DHS revised unmet need framework (v626a), clarify the restriction to currently pregnant women, and improve methodological transparency.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

What do you mean by initial association? What are the possible confounders and why? Why are we doing this? Why do we need variable selection? How many models do we have and why? Earlier it was 0.25. Please justify. Why are you repeating things again and again? Can we interpret these values? Why do you need to mention them in the methods section? (AIC, BIC, model comparisons, and repeated values)

Author Response (Page 9-10, Lines 232-234, and 238-246)

We sincerely thank the reviewer for this detailed and important methodological feedback.

In the revised manuscript, initial association revised into crude association and we have substantially streamlined the statistical analysis section to improve readability and to align with standard reporting practices for observational studies. Specifically, we have updated the detailed stepwise description of multiple intermediate models, repeated presentation of AIC/BIC values, and unnecessary reporting of model comparison results in the Methods section.

We now clarify that variable selection was based on a combination of theoretical relevance and bivariate screening ($p < 0.25$), which is a commonly used and widely accepted threshold to avoid premature exclusion of potential confounders. All covariates were selected a priori based on prior literature and epidemiological plausibility, and were included in a single survey-weighted multivariable logistic regression model.

Model diagnostics, including multicollinearity assessment and goodness-of-fit testing, are now briefly reported without iterative model comparisons. Detailed model comparison results (if required) have been removed from the Methods section to avoid redundancy and are reported only where appropriate in the Results section.

We believe these revisions substantially improve clarity, reduce repetition, and ensure that the Methods section reflects standard epidemiological reporting conventions.

Changes made:

The statistical analysis section has been rewritten to remove repeated descriptions of model building steps and AIC/BIC comparisons, retain only essential variable selection criteria, and improve alignment with STROBE reporting guidelines.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

Why are you using $p < 0.25$ and $p < 0.05$? Please justify.

Author Response (Page 10, Lines 240-246)

We sincerely thank the reviewer for this important methodological clarification. In the revised manuscript, we clarify that the two thresholds serve different purposes in the analytical framework. The $p < 0.25$ threshold in the bivariate analysis was used as a liberal screening criterion to identify potential candidate variables for multivariable modeling. This approach is commonly recommended in epidemiological and regression modeling literature to reduce the risk of excluding variables that may become significant or act as confounders in the adjusted model.

In contrast, the $p < 0.05$ threshold was used in the final multivariable logistic regression model to determine statistical significance of associations after adjusting for all selected covariates. This represents the conventional level of statistical significance used for hypothesis testing in inferential analysis.

We have revised the Methods section to clearly distinguish between these two stages and to improve transparency in the model-building strategy.

Changes made:

The Statistical Analysis section has been updated to explicitly distinguish between the variable screening criterion ($p < 0.25$) used in bivariate analysis and the statistical significance level ($p < 0.05$) used in the final adjusted model, and to clarify their distinct purposes in confounder selection and inference.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

Why do we need confounder selection?

Author Response (Page 10, Lines 242-246)

We sincerely thank the reviewer for this important methodological question. Confounder selection is essential in observational studies to obtain unbiased estimates of the association between internet use and unmet need for family planning, as randomization is not possible in this study design. Adjustment for confounding variables helps to control for factors that may influence both exposure and outcome, thereby reducing potential bias in the estimated associations.

In this study, potential confounders were identified a priori based on existing literature, epidemiological plausibility, and guided by a conceptual framework presented in Figure 2 (DAG). The directed acyclic graph (DAG) was used to inform the selection of covariates included in the multivariable model, ensuring that only theoretically justified confounders were adjusted for in the final analysis.

Changes made:

The rationale for confounder selection has been clarified in the Methods section, and explicit reference to the conceptual framework (Figure 2, DAG) has been added to strengthen the theoretical justification for covariate adjustment.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

Why are AIC and BIC reported repeatedly and used again and again in the manuscript?

Author Response

We sincerely thank the reviewer for this important methodological observation.

In the revised version, we have substantially simplified the presentation of model selection procedures. Specifically, we have removed repeated reporting of AIC and BIC values from intermediate steps and eliminated their iterative discussion within the Methods section. Instead, we now briefly state that AIC and BIC were used only as supplementary criteria during internal model development to support selection of a parsimonious final model.

The final presentation focuses on the adjusted multivariable logistic regression results, while model-building details have been streamlined to avoid redundancy and improve readability in accordance with standard epidemiological reporting guidelines.

Changes made:

All repeated references to AIC and BIC across stepwise model building have been removed or consolidated, and their use is now briefly mentioned only once as part of model selection criteria in the Methods section.

RESPONSE TO REVIEWER 1 – COMMENT

Reviewer Comment:

Is this for national/rural/urban? Why do you have only one value in the cOR? Crude is for national.

Author Response (Page 16, line no: 334-337, Table 2)

We sincerely thank the reviewer for this important clarification. We agree that the presentation of crude odds ratios (cOR) in Table 2 required clearer specification to avoid ambiguity.

In the revised manuscript, we have clarified that the crude odds ratio (cOR) represents the unadjusted association estimated for the overall national sample, whereas the adjusted odds ratios (aORs) are presented separately for the national, rural, and urban subgroups to assess potential effect modification by place of residence.

Thus, the cOR is reported only once because it reflects the overall unadjusted association between internet use and unmet need for family planning in the full analytic sample (N = 4,370). In contrast, stratified adjusted models (rural and urban) were estimated separately to explore heterogeneity in the association across residence types.

We have revised Table 2 to explicitly label the cOR as “National (unadjusted)” and clearly distinguish it from the stratified adjusted models to improve interpretability and avoid confusion.

Changes made:

Table 2 has been updated with clearer labeling indicating that cOR corresponds to the national-level unadjusted model, while aORs are reported for national and stratified (rural and urban) survey-weighted adjusted models.

Reviewer Comment:

Is this from the same analysis as Table 2?

Author Response: (page no:17, Table 3)

We sincerely thank the reviewer for this important clarification. Table 2 and Table 3 are based on different analytical objectives and therefore come from separate modeling strategies. Table 2 presents the primary crude (cOR) and fully adjusted (aOR) associations between internet use and unmet need for family planning using the overall national sample, which constitutes the main inferential model of the study.

In contrast, Table 3 reports stratified analyses conducted to explore effect heterogeneity across key socio-demographic subgroups, including maternal age, age at first birth, division, maternal education, and household media exposure. For these analyses, separate multivariable logistic regression models were fitted within each subgroup rather than extending the full national model. The aim was not to compare effect sizes directly with Table 2 but to assess consistency and variation of associations across population strata.

Accordingly, Table 3 should be interpreted as subgroup-specific adjusted associations rather than an extension of the main model presented in Table 2. This distinction has now been clarified in the Methods and Table 3 footnote to avoid confusion.

REVIEWER 2

RESPONSE TO REVIEWER 2 – COMMENT 1

Reviewer Comment:

Title and the Abstract indicate that the paper addresses ever married women but Methods section in the body of the manuscript describes the sample participants are currently pregnant women. I'm wondering if it used a sample of ever married pregnant women. If so, I'd indicate that.

Author Response (Title, Abstract, Methods section; Page 1, Lines 1-2, page no:2, line no: 32, 36,42, page no: 3-4, line no: 83-85, page no: 5, line no: 121-122, page no: 6, line no: 157-158, page 7-8 line no: 168-177)

We sincerely thank the reviewer for this important observation. We agree that the earlier version of the manuscript may have led to confusion regarding the study population due to inconsistent terminology across sections. In response to this comment, we have carefully revised the manuscript to ensure full consistency in the description of the analytical sample. The study population has now been clearly specified as currently pregnant women aged 15–49 years using data from the Bangladesh Demographic and Health Survey (BDHS) 2022. Accordingly, the title of the manuscript has been revised to reflect this population accurately, and all references to “ever-married women” have been removed or corrected throughout the Abstract and Methods sections. Furthermore, we have clarified the eligibility criteria and sample selection process in the Methods section to ensure that the analytical framework is transparent and consistent with the revised study population. These revisions have been made to improve clarity, avoid misinterpretation, and ensure alignment between the study title, abstract, and analytical methods.

Changes made:

Revised the title to specify currently pregnant women; updated the Abstract and Methods sections for consistency; and clarified the eligibility criteria and analytical sample description throughout the manuscript.

RESPONSE TO REVIEWER 2 – COMMENT 2

Reviewer Comment:

Figure 1 is most uninformative. I'd create a sample flowchart clearly showing the numbers of included and excluded participants at each stage, and the missing values, ultimately reaching the final analytical sample.

Author Response (Page 7, line no: 161-162, Figure 1)

We sincerely thank the reviewer for this valuable suggestion. We agree that the clarity of the participant selection process is essential for transparency and reproducibility, particularly in secondary analyses using large survey datasets such as the BDHS 2022. In response to this comment, we have substantially revised Figure 1 to improve the presentation of the study sample selection process.

The revised figure now provides a clear stepwise flow diagram showing the derivation of the final analytical sample from the original BDHS dataset. It explicitly presents the number of eligible respondents at each stage, the exclusions due to missing values and non-eligibility criteria, and the final number included in the analysis. This revision ensures that the selection process is fully transparent and allows readers to clearly understand how the final study population was derived from the original survey sample.

Changes made:

Figure 1 has been completely redesigned to present a structured flow of sample selection, including all exclusions due to missing data and eligibility criteria, leading to the final analytical sample.

RESPONSE TO REVIEWER 2 – COMMENT 3

Reviewer Comment:

I'm curious about how the authors addressed the cluster nature of the data. You might consider using mixed or multilevel models, or adjust for the cluster variable in current models.

Author Response (Page 9-10, Lines 231--234, 2.6. Survey Design and Weighting)

We sincerely thank the reviewer for this important methodological comment.

In the present study, this survey design structure was fully taken into account in the statistical analysis. Specifically, the analysis already incorporated the clustering effect by applying survey-weighted logistic regression using the “svy” command in Stata. This approach adjusts standard errors for intra-cluster correlation and ensures unbiased variance estimation under the complex sampling design of BDHS. In addition, sampling weights and stratification were also applied to produce nationally representative estimates.

This analytical strategy is consistent with the DHS methodological guidelines and widely adopted in peer-reviewed studies using BDHS data, where survey-adjusted regression models are preferred for population-level inference. Therefore, the clustering nature of the data has been appropriately addressed within the current modelling framework, and no additional modelling change was required.

To improve transparency, we have now explicitly clarified this procedure in the Statistical Analysis section of the revised manuscript.

Changes made:

The Statistical Analysis section has been updated to explicitly state that survey weights, clustering (PSU), and stratification were incorporated using survey-weighted logistic regression (svy: logistic) in Stata, in accordance with BDHS analytical standards.

RESPONSE TO REVIEWER 2 – COMMENT 4

Reviewer Comment:

For model testing, instead of citing AUC value of .78, readers expect to know if this model performs better than an alternative or at a minimum the null model. Also, show the model validation statistics in each table and cite the name of the survey and sample size in a footnote so that it can stand alone irrespective of the text.

Author Response (Page 31, Tables 4, Lines 697-704)

We sincerely thank the reviewer for this insightful comment regarding model evaluation and reporting clarity. We fully agree that model performance should not rely solely on a single discrimination metric such as the AUC, and instead should be assessed through appropriate comparisons with baseline and alternative models.

In response to this comment, we have revised the model evaluation strategy in the manuscript to provide a more comprehensive assessment of model fit. Specifically, we now compare the null model (M0) with successive model specifications, culminating in the final adjusted model (M14). Model improvement is evaluated using likelihood ratio tests, as well as information criteria including Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). These results clearly demonstrate that the final model (M14) provides the best overall fit among all fitted models.

In addition, we have ensured that all regression tables are self-contained by including the survey name (Bangladesh Demographic and Health Survey 2022) and the analytical sample size (N = 4,370) in the table footnotes. We have also explicitly stated that all analyses were conducted using survey-weighted logistic regression accounting for sampling weights, clustering, and stratification to reflect the complex survey design.

The AUC value is now retained only as a supplementary measure of model discrimination and is interpreted alongside likelihood-based model fit statistics rather than as the primary indicator of model performance.

Changes made:

Model evaluation has been expanded to include comparison of the null model (M0) and final model (M14) using likelihood ratio tests, AIC, and BIC. All tables have been updated to include BDHS 2022 survey information and sample size in footnotes to ensure standalone interpretability. The role of AUC has been revised as a supplementary diagnostic measure rather than the primary model evaluation criterion.

RESPONSE TO REVIEWER 2 – COMMENT 5

Reviewer Comment:

The authors report a very high level of unmet need compared to that in the BDHS 2022 report. Is it because of highlighting a different sample? I'd explain this difference for clarity.

Author Response (Page 6, Lines 157-158)

We sincerely thank the reviewer for this important and insightful comment. We agree that the observed difference in the prevalence of unmet need for family planning compared to the BDHS 2022 national report requires clear clarification.

The higher prevalence observed in our study is primarily attributable to differences in the study population and analytical definition compared to the standard BDHS 2022 estimates. The DHS national report estimates unmet need among all currently married women aged 15–49 years using the full survey sample. In contrast, our study specifically restricted the analysis to currently pregnant women aged 15–49 years, resulting in a more selective and biologically distinct subpopulation.

Since currently pregnant women represent a subgroup with different fertility status and contraceptive dynamics compared to the general population of married women, the prevalence of unmet need is expected to differ from the national estimate reported in the BDHS final report. This distinction has now been explicitly clarified in the revised Results and Discussion sections to avoid misinterpretation and to ensure appropriate comparison with existing literature.

Changes made:

We have added a clear explanation in the Results and Discussion sections clarifying that the study estimates are based on a restricted sample of currently pregnant women, whereas BDHS national estimates are based on all currently married women, which accounts for differences in reported prevalence.

RESPONSE TO REVIEWER 2 – COMMENT 6

Reviewer Comment:

Please share the code for transparency and reproducibility.

Author Response (Page 23, Lines 470-472, 4.8 Codes Availability)

We sincerely thank the reviewer for emphasizing the importance of transparency and reproducibility. In response to this comment, we have made the full analysis code publicly available to ensure reproducibility of all study findings.

The Stata code used for data cleaning, variable construction, and statistical analysis has been deposited in a publicly accessible GitHub repository. The repository contains all scripts necessary

to replicate the analytical procedures reported in the manuscript, including data preprocessing, model building, and survey-weighted regression analyses.

The code availability statement has been updated in the “Code Availability” section of the revised manuscript to include the repository link.

Changes made:

A public GitHub repository containing all analysis code has been provided, and the manuscript has been updated to include a formal code availability statement for transparency and reproducibility.

RESPONSE TO REVIEWER 2 – COMMENT 7

Reviewer Comment:

The text below sounds vain, grammatically incorrect and AI generated. Please use clear language and avoid using the term, generative health.

Author Response

We sincerely thank the reviewer for this valuable suggestion regarding clarity and language quality. We agree that the original sentence contained unclear phrasing and terminology that could affect readability and scientific precision.

In response, we have completely revised the sentence to improve clarity, grammatical correctness, and academic tone. The term “generative health” has been removed, and the statement has been rewritten in clear and concise language consistent with standard scientific writing.

The revised sentence now reads as follows:

“Understanding how internet use associate’s reproductive health outcomes may help inform targeted interventions aimed at reducing unmet need for family planning among women in Bangladesh.”

Changes made:

The original sentence containing unclear and inappropriate terminology has been removed and replaced with a clear, concise, and academically appropriate statement.

RESPONSE TO REVIEWER 2 – COMMENT 7

Reviewer Comment:

The text below sounds vain, grammatically incorrect and AI generated. Please use clear language and avoid using the term, generative health.

Author Response

We sincerely thank the reviewer for this valuable suggestion regarding clarity and language quality. We agree that the original sentence contained unclear phrasing and terminology that could affect readability and scientific precision.

In response, we have completely revised the sentence to improve clarity, grammatical correctness, and academic tone. The term “generative health” has been removed, and the statement has been rewritten in clear and concise language consistent with standard scientific writing.

The revised sentence now reads as follows:

“Understanding how internet use influences reproductive health outcomes may help inform targeted interventions aimed at reducing unmet need for family planning among women in Bangladesh.”

Changes made:

The original sentence containing unclear and inappropriate terminology has been removed and replaced with a clear, concise, and academically appropriate statement.

RESPONSE TO REVIEWER 2 – COMMENT 8

Reviewer Comment:

The phrase, 'the national association between internet use and unmet need' is awkward. Using the national level data does not make an association national.

Author Response (Page 4-5, Lines 114-116)

We sincerely thank the reviewer for this helpful suggestion. We agree that the original phrasing was not appropriate, as the term “national association” may be misleading in statistical interpretation.

In response, we have revised the manuscript to replace this expression with more accurate terminology. The revised text now clearly indicates that the study assesses the association between internet use and unmet need for family planning using nationally representative data from the Bangladesh Demographic and Health Survey (BDHS) 2022. This revision improves conceptual clarity and aligns with standard epidemiological reporting practices.

Changes made: The phrase “national association” has been removed and replaced with “association using nationally representative BDHS 2022 data” throughout the manuscript.

RESPONSE TO REVIEWER 2 – COMMENT 9

Reviewer Comment:

Please share a supplemental table, showing separate columns for each variable's name, question, responses, and coding scheme, for transparency.

Author Response (Page 9, **Supplementary Table S1**, Lines 219)

We sincerely thank the reviewer for this valuable suggestion regarding transparency and reproducibility. In response to this comment, we have developed and included a comprehensive supplementary table (**Supplementary Table S1**) that details all study variables used in the analysis. The supplementary table clearly presents each variable's name, corresponding DHS code, survey question or description, original response categories, and final recoding scheme used in the statistical analysis. This ensures full transparency in variable construction and allows readers to clearly understand how each analytical variable was derived from the Bangladesh Demographic and Health Survey (BDHS 2022) dataset.

We have ensured that all transformations, including recoding of categorical variables, construction of composite indices, and creation of binary outcomes, are fully documented in this table. This addition substantially improves the reproducibility of the study and aligns the manuscript with standard reporting requirements for secondary data analysis using DHS data.

Changes made:

A new **Supplementary Table S1** has been added, providing detailed information on variable definitions, survey questions, original response categories, and final coding schemes for all analytical variables used in the study.